

Profile

Machine learning engineer and PhD researcher in reinforcement learning, with experience building Python research software, benchmarking infrastructure, and scalable learning algorithms. Strong background in PyTorch/JAX, sequential decision-making, optimization, and large-scale experimentation. Interested in open-source ML tooling, efficient training/inference, and developer-facing ML libraries.

Experience

Oct. 2025 – Mar. 2026 **Quantitative Researcher, Qube Research & Technologies, Paris**

- Developed machine-learning and representation-learning methods on large-scale datasets covering more than 3,000 equity assets.
- Designed robust experimentation and validation frameworks for predictive modeling, including feature engineering, out-of-sample evaluation, and statistical performance analysis.
- Built scalable Python/JAX research infrastructure for training, benchmarking, and evaluating learning-based decision systems under noisy real-world data.

2022 – 2026 **PhD Researcher in Reinforcement Learning, CMAP, École Polytechnique**

- Developed state-abstraction and progressive-disaggregation algorithms for large-scale Markov decision processes, reducing the effective planning space while preserving control-relevant structure.
- Demonstrated consistent speedups on benchmark and applied MDPs, including random MDPs, Four Rooms, Mountain Car, Sutton's racetrack, tandem queues, and hydro-valley management models.
- Built reusable Python/Gymnasium benchmarking infrastructure for RL agents and dynamic-programming solvers, including experiment configuration, evaluation protocols, and reproducible benchmark suites.

2021 – 2024 **Research Collaboration in RL and Combinatorial Optimization, Orange Gardens / CMAP**

- Developed RL and optimization methods for latency-constrained service placement in edge-computing systems.
- Benchmarked exact, heuristic, and learning-based approaches in an industrial 6G research context, leading to an IFIP Networking 2025 publication.

Selected Publications

- 2025 Orso Forghieri, ***Hierarchical Reinforcement Learning for Large Scale Problems***, PhD thesis, Institut Polytechnique de Paris.
- 2025 Orso Forghieri, Yannick Carlinet, Emmanuel Hyon, Erwan Le Pennec and Nancy Perrot, ***Faster Latency Constrained Service Placement in Edge Computing with Deep Reinforcement Learning***, *IFIP Networking*, 2025.
- 2024 Orso Forghieri, Hind Castel, Emmanuel Hyon and Erwan Le Pennec, ***State Abstraction Discovery from Progressive Disaggregation Methods***, *European Workshop on Reinforcement Learning*, 2024.
- 2024 Orso Forghieri, Hind Castel, Emmanuel Hyon and Erwan Le Pennec, ***Progressive State Space Disaggregation for Infinite Horizon Dynamic Programming***, *International Conference on Automated Planning and Scheduling*, 2024.

Education

- Sep. 2022 – Dec. 2025 **PhD in Applied Mathematics / Reinforcement Learning**, École Polytechnique, Institut Polytechnique de Paris.
Thesis: *Hierarchical Reinforcement Learning for Large Scale Problems*.
Supervision: Erwan Le Pennec, Hind Castel-Taleb, Emmanuel Hyon.
Laboratory: Centre de Mathématiques Appliquées, École Polytechnique.
- 2021 – 2022 **MSc MVA – Mathematics, Vision, Learning**, ENS Paris-Saclay.
Focus: statistical learning, deep learning, reinforcement learning, time series analysis, and optimization.
Thesis: *Hierarchical Reinforcement Learning for Optimizing Large Systems*, supervised by Hind Castel-Taleb and Emmanuel Hyon.
- 2018 – 2021 **Engineering Degree**, École Polytechnique.
Coursework: statistical learning, probability, optimization, algorithms, deep learning, uncertainty quantification, risk analysis.
Research project: *An Overview of Market Forecast with Machine Learning Techniques*.

Open-Source Projects

- GitHub **Research Software for Reinforcement Learning and Dynamic Programming**
- Developed open-source Python implementations of state-abstraction, aggregation/disaggregation, and dynamic-programming algorithms for large-scale Markov Decision Processes.
 - Built reusable benchmarking frameworks and Gymnasium-compatible environments for evaluating planning and reinforcement-learning methods across control, queueing, and resource-allocation problems.
 - Released research code accompanying peer-reviewed publications (ICAPS 2024, EWRL 2024), with emphasis on reproducibility, scalable experimentation, and algorithmic evaluation.
 - Maintained personal research software and documentation through public GitHub repositories and a research website.

Technical Skills

- Frameworks PyTorch, JAX, TensorFlow, Gymnasium/Gym, Hugging Face Transformers, Datasets, Accelerate
- ML / AI Deep Learning, Reinforcement Learning, Representation Learning, Sequential Decision-Making, Large-Scale Experimentation, Statistical Learning
- Systems Python, C++, SQL, Linux, Git, Testing, Benchmarking, Reproducible ML Pipelines
- Optimization Dynamic Programming, Stochastic Optimization, Combinatorial Optimization, Integer Programming, Approximate Planning